$$\begin{bmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 3 & 6 & 8 & 10 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & \boxed{2} & 4 \\ 0 & 0 & 2 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 & -2 \\ 0 & 0 & \boxed{2} & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

U

need 1.

so divide whole row by 2.

$$\begin{bmatrix} 1 & 2 & 0 & -2 \\ 0 & 0 & 2 & \end{bmatrix}$$

$\Rightarrow$ rref:

$$\begin{bmatrix} \boxed{1} & 2 & 0 & -2 \\ 0 & 0 & \boxed{1} & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow$$

ignore.

R

$$c\begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + d\begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}$$

$$F = \begin{bmatrix} -2 & 2 \\ 1 & 0 \\ 0 & -2 \end{bmatrix} \quad F = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$$

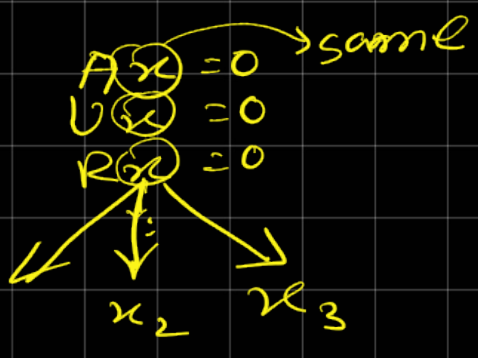
$$F = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$R = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix} \begin{bmatrix} | & | \\ | & | \end{bmatrix} \rightarrow$$ columns are solutions to $Ax = 0$.

L_{RN} $\Sigma_{v_1 v_2 v_3}$ N

$RN = 0$?

$\begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix} \begin{bmatrix} -F \\ I \end{bmatrix} = 0$ $\quad -F + F = 0$ $\quad \therefore F = -F.$



$\begin{bmatrix} | & | & | \\ x_1 & x_2 & x_3 \\ | & | & | \end{bmatrix}$

$N \rightarrow$ ⟶ Mid

Final:

Complete Soln of $Ax = b$

$$\left[\begin{array}{cccc:c} 1 & 2 & 2 & 2 & b_1 \\ 2 & 4 & 6 & 8 & b_2 \\ 3 & 6 & 8 & 10 & b_3 \end{array}\right] \begin{matrix} b_1 \\ b_2 - 2b_1 \\ b_3 - 3b_1 \end{matrix}$$

$\Rightarrow b_3 - b_2 - b_1 = 0$

b1 will there be solutions?

b2-2b1 How many?

For which right hand side b is $Ax = b$ solvable?

otherwise no matter how many combinations I take, b won't be in the col-space.

$\begin{bmatrix} \quad \end{bmatrix} \begin{bmatrix} x \end{bmatrix}$

if and only if b is in col-space of A.

(cz linear combination of A)

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$x\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + y\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + z\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$\Rightarrow b_3 - b_1 - b_2 = 0$ (cz 1 row is all zeros)

$\Rightarrow b_3 = b_2 + b_1 \Longrightarrow$ cond^{n} for solvability.

$$\left[\begin{array}{cccc|c} 1 & 2 & 2 & 2 & 5 \\ 2 & 4 & 6 & 8 & 1 \\ 3 & 6 & 8 & 10 & c \end{array}\right]$$

$Ax = b$

$Ax = b$

particular $\rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

$Ax = 0$

$$L x \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} + d \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix}$$

compute $Ax = b \rightarrow$ $x_p + x_n$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + c \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} + d \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix}$$

x_p

$Ax_p + Ax_n = b \rightarrow$ gives 0 so gotta add null space.

x has particular soln & nullspace.

$A(x_p + x_n) = b$

$5(-3) + 6 = -9$

$$\left[\begin{array}{cccc|c} 1 & 2 & 2 & 2 & 5 \\ 0 & 0 & 2 & 4 & -9 \\ 0 & 0 & 0 & 0 & -9 \end{array}\right]$$

$0 = -9$ ✗

$$\left[\begin{array}{cccc|c} 1 & 2 & 2 & 2 & 5 \\ 0 & 0 & 2 & 4 & -9 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

x_2 x_4 — free variables

$2x_3 + 4x_4 = -9$ → $x_3 = -\frac{9}{2}$

$x_1 + 2x_2 + 2x_3 + 2x_4 = 5$

$x_1 + 2(-9/2) = 5$

$x_1 = 14$

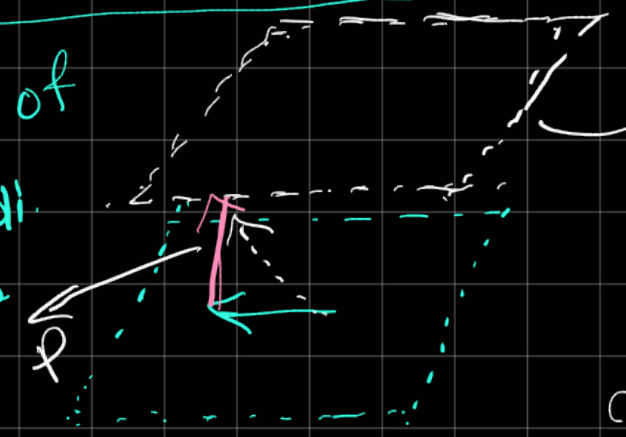
The free col can be ignored as they r independent col so, they r 0

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 14 \\ 0 \\ -9/2 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 14 \\ 0 \\ -9/2 \\ 0 \end{bmatrix} + c\begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + d\begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

x_p x_n

the



not part of nullspace if it was the multiplication would give b.

escapes origin, so not vector space. vector set

as $Ax=b$

so adding any vector to a vector from null space, so it will escape nullspace.

(1) $r = \#$ no of pivots. $m \times n$

$r = n = \text{col}$

→ all the cols have pivots.

$$\begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 5 & 1 \end{bmatrix}$$ how many solutions?

↓ rref.

$$\Rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} I \\ 0 \end{bmatrix}$$

either 1 or 0 solutions.

(2) $r = m$, $r \leq n$

all rows with single pivots.

$$\begin{bmatrix} 1 & 2 & 3 & 5 \\ 1 & 1 & 1 & 1 \end{bmatrix} \overset{rref}{=} \begin{bmatrix} I & F \end{bmatrix}$$

∞ Full rank.

③ $r = n = m$

$[\ f\ ]$ only one solution.

④ $r < n$, $r < m$

$$\begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}$$

0 or ∞ solutions.